STALKS OF THE PUSHFORWARD ALONG A CLOSED IMMERSION OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a three-step reduction to the scheme case; the base change formula for the pushforward is what makes the replacement legitimate.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The base change formula for the small pushforward along an etale morphism. Properties of Spaces, Lemma spaces-properties-lemma-pushforward-etale-base-change, used at proof lines 2628-2630. Allows the replacement of X by an etale neighbourhood U.
- The scheme case for stalks of a pushforward along a closed immersion. Etale Cohomology, Lemma etale-cohomology-lemma-stalk-pushforward-closed-immersion, used at proof lines 2631-2633. Is the input that closes the computation.
- Stalks computed after restriction to an etale neighbourhood. used at proof lines 2625-2627. Justifies computing the stalk at the chosen geometric point of U.

Source reference. Original lemma 04FX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5813–5813. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03LR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4723–4723. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $i : Z \to X$ be a closed immersion of algebraic spaces over S. Let $\overline{z}$ be a geometric point of Z with image $\overline{x}$ in X. Then $(i_{small,*}\mathcal{F})_{\overline{z}} = \mathcal{F}_{\overline{x}}$ for any sheaf $\mathcal{F}$ on $Z_{étale}$.*

Proof. Choose an étale neighbourhood $(U, \overline{u})$ of $\overline{x}$. Then the stalk $(i_{small,*}\mathcal{F})_{\overline{z}}$ is the stalk of $i_{small,*}\mathcal{F}|_U$ at $\overline{u}$. By Properties of Spaces, Lemma 03LR we may replace X by U and Z by $Z \times_X U$. Then $Z \to X$ is a closed immersion of schemes and the result is Étale Cohomology, Lemma 04FX. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.